Mapping the History of European Nuclear Fusion Cooperation: Combining Archival and Publication Data

Malte Vogl*†1 and Lalli Roberto*‡2

[1]Max Planck Institute of Geoanthropology – Germany
[2]Politecnico di Torino = Polytechnic of Turin (Polito) – Italy

Abstract

Historians of European integration and nuclear energy have examined fusion research as a site where big science, Cold War politics, and international community-building intersect. Yet, even in this growing literature, the development of cooperation between scientists, administrators, and policy-makers tends to be described qualitatively, and systematic analyses of the social networks underpinning fusion research have not been pursued. Within the project FusEUrope-European cooperation in nuclear fusion research: from history to future policy design, we address this gap by investigating the history of European techno-scientific cooperation in nuclear fusion and its relationship to political integration from the mid-1950s to the early 2000s
(https://fuseurope.polito.it/). One objective is to reconstruct longitudinal social networks of actors and institutions, drawing on the socio-epistemic network framework (Lalli et al. 2020, Kaye et al. 2024)

In this paper we present first results and open challenges of this study especially for what concerns the reconstruction of the social multi-layer of these networks based on two kinds of sources: 1) archival records from European and national bodies for building a database on committee memberships, association contracts, project boards, and advisory groups (2) publication metadata from bibliographic databases, used to derive co-authorship and institutional collaboration ties. We present temporal multi-layer networks of persons, institutions and countries (Vogl & Lalli 2025), highlight the limits and possibilities of comparing and mixing publication repositories (Web of Science and OpenAlex) and archival records for research in the history of techno-scientific development and discuss key elements of the social clustering processes leading to reconfigurations of the networks topology and its reflection in language metrics (Schlattmann & Vogl 2025).

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*Speaker
†Corresponding author: vogl@gea.mpg.de
‡Corresponding author: roberto.lalli@polito.it

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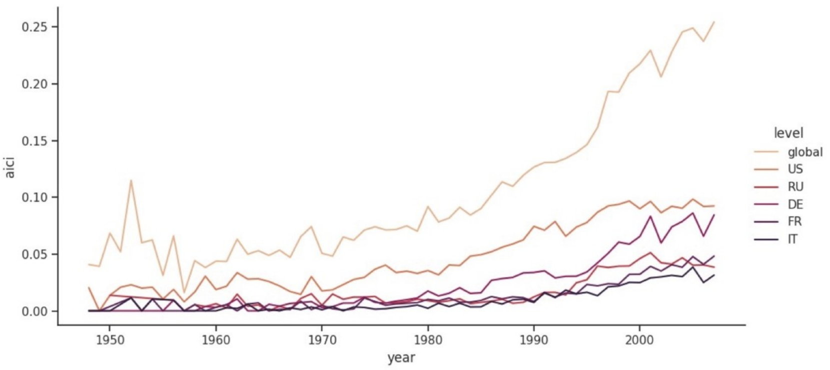

Figure 1: Adjusted international collaboration index between 1950 and 2007 for all publications in OpenAlex related to fusion research journals or topics (N~220.000) and for specific countries (US, Russia, Germany, France and Italy). The index expresses the amount of publications with international affiliations in relation to available metadata (Vogl & Lalli 2025).

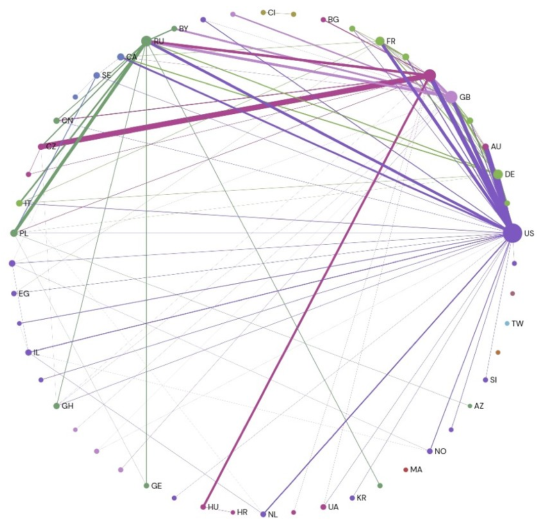

Figure 2: Co-country network for publications between 1974-1976. Node size given by degree, edge width by weight. Color set by Louvain community detection. The plot highlights the inter-mediator roles of Japan and Germany in the fusion research community. For data generation see (Vogl & Lalli 2025). Visualized with Gephi Lite.